Semantically-Structured Multimodal 3D Descriptor Learning for Urban Place Recognition

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***Abstract*—Understanding environments has increasingly become a critical task for autonomous vehicle navigation applications in highly dynamic scenarios. Although extracting information from images has shown promising results, it remains challenging under changing environmental conditions, not only due to scene dynamics but also variations in illumination. On the other hand, point clouds acquired from LiDAR sensors are largely unaffected by weather conditions; however, they still present challenges related to high data sparsity and the lack of contextual information in perception tasks. Therefore, semantic understanding of the environment is essential and remains an open challenge, particularly when fusion mechanisms are involved. Based on this motivation, this paper proposes a multimodal semantic feature extraction strategy based on a fused representation of panoramic imagery and geometrically consistent projections. Unlike approaches that focus solely on semantic prediction, the proposed method explicitly derives compact and discriminative semantic descriptors from the fused representation. These descriptors jointly encode visual appearance, contextual information, and spatial structure, enabling robust object-level class identification in complex scenarios. Experiments are conducted in KITTI Dataset.**

***Index Terms*—Camera, 3D-LiDAR, sensor fusion, semantic descriptor, multi-modal fusion.**

I. Introduction

Place recognition is a fundamental problem in autonomous navigation and mobile robotics, as it enables loop closure detection, global localization, and long-term map consistency. The goal is to generate compact yet highly discriminative scene descriptors that uniquely characterize a location while remaining invariant to viewpoint changes, illumination variations, seasonal effects, and dynamic objects. Designing such robust representations remains a central challenge in perception systems. Early place recognition approaches relied on handcrafted visual descriptors such as SIFT [1], [2], SURF [3], [4], ORB [5], HOG [6]. The introduction of deep learning significantly improved robustness through convolutional architectures capable of learning hierarchical semantic representations.

A seminal contribution in this direction was NetVLAD [7], which proposed a differentiable global descriptor aggregation layer enabling end-to-end training for large-scale place recognition. Although image-based global descriptors benefit from dense texture and strong contextual reasoning, they remain highly sensitive to illumination changes, shadows, weather conditions, and perceptual aliasing, where visually similar scenes correspond to distinct physical locations.

To overcome the limitations of purely appearance-based methods, LiDAR-based place recognition techniques operate directly on 3D point clouds, leveraging geometric structure instead of texture. Working in 3D space provides improved invariance to lighting conditions and enables more accurate discrimination of scene elements based on spatial configuration. The emergence of learning-based architectures such as PointNet [8] and PointNet++ [9] enabled direct feature extraction from unordered point sets through permutation-invariant operations and hierarchical neighborhood aggregation. Building upon these foundations, several works specifically targeted global 3D place recognition. PointNetVLAD [10] combined PointNet-based feature extraction with a NetVLAD aggregation layer to learn discriminative global descriptors from raw point clouds. OverlapNet [11] proposed a LiDAR-based method that estimates overlap and relative yaw angle between scans, improving robustness in loop closure detection. More recently, MinkLoc3D [12] leveraged sparse 3D convolutions within the Minkowski Engine framework to generate highly discriminative global descriptors directly from voxelized point clouds. These approaches demonstrated that learned 3D geometric descriptors significantly outperform handcrafted methods in large-scale outdoor localization.

In addition to global representations, learning-based local geometric descriptors such as FCGF (Fully Convolutional Geometric Features) [13] and D3Feat [14] have shown strong performance in point cloud matching and registration tasks. These models exploit sparse convolutional [15] architectures and joint detector–descriptor learning [16] to enhance distinctiveness and repeatability in 3D space. Despite these advances, relying exclusively on LiDAR data presents inherent challenges. Point clouds are sparse, irregular, and range-dependent, leading to uneven point distributions and reduced resolution at long distances. This dispersion weakens contextual modeling and limits semantic abstraction. While 3D geometry captures structural consistency, it often lacks appearance cues that help disambiguate perceptually similar environments. Conversely, purely image-based descriptors provide rich texture and semantic information but suffer from sensitivity to illumination, weather, and dynamic elements.

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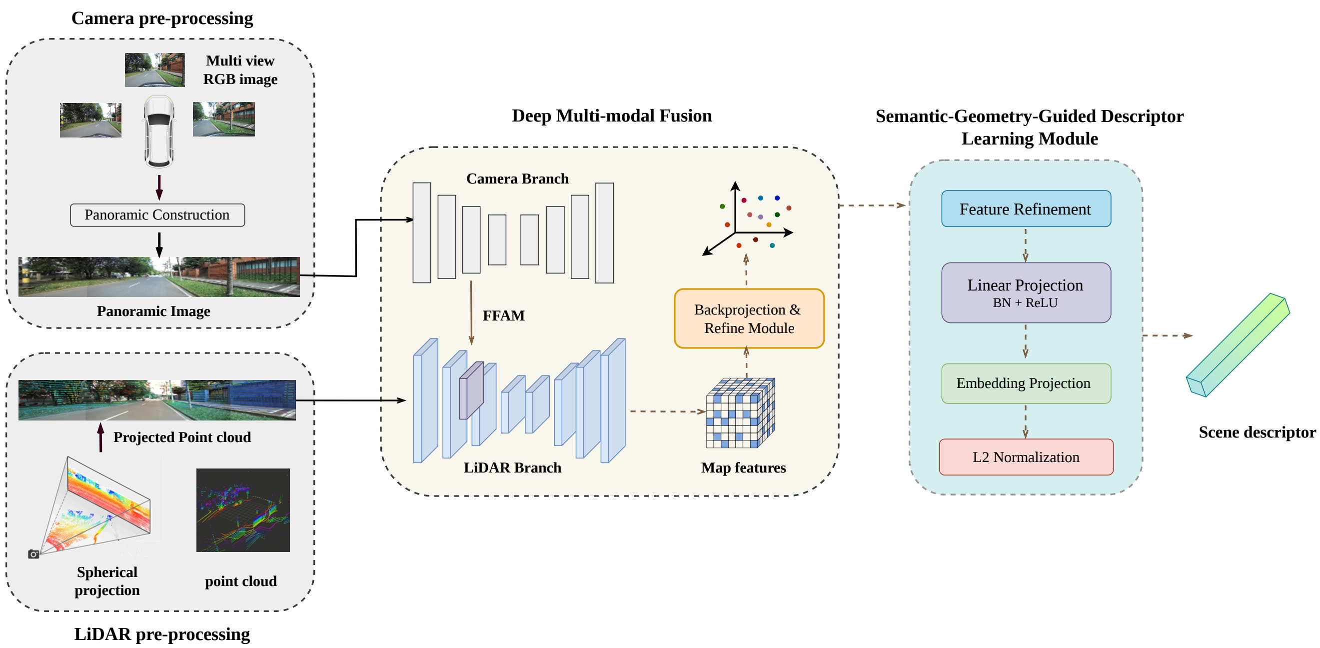

Fig. 1: Main core of the proposed pipeline for generating global descriptors from multimodal data. The point cloud is projected onto the panoramic image, and both modalities are processed through a shared backbone to extract refined 3D features. A learnable global descriptor head aggregates these features into a compact embedding for place recognition.

Large-scale annotated datasets [17], [18], [19], [20] have highlighted the importance of semantic supervision for structured scene understanding; however, extracting semantically robust and globally discriminative representations from single modalities remains challenging. These observations motivate multimodal fusion strategies that integrate LiDAR geometry with visual appearance. By leveraging the complementary nature of spatial structure and texture, multimodal approaches mitigate the modality-specific degradation factors that affect single-sensor systems. Nevertheless, multimodal fusion introduces additional challenges, including precise extrinsic calibration, accurate projection of 3D points onto the image plane, reliable cross-modal alignment, and increased computational complexity.

Motivated by the approach presented in [21], our pipeline as depicted in Fig 1 leverages panoramic images generated through a homography-based mosaicing of three stereo cameras, together with the corresponding LiDAR point cloud. The 3D measurements are projected onto the panoramic mosaic, effectively transforming geometric information into a 2D representation aligned with the image domain. In this manner, each pixel is enriched with geometric attributes associated with a spatial point prior to being processed by the semantic segmentation network. The output of the point cloud branch is subsequently re-projected into the 3D domain, following the strategy introduced in [22], In addition to semantic segmentation, we introduce a learnable global descriptor head that transforms refined 3D multimodal features into a compact and discriminative embedding suitable for place recognition. Unlike conventional approaches that rely on handcrafted pooling or intermediate logits, the proposed descriptor is extracted directly from the refined 3D feature space and aggregated through an attention-based global pooling mechanism. This design allows the network to selectively emphasize structurally stable and semantically informative regions while suppressing dynamic or less discriminative elements.

To validate our framework, we conducted extensive experiments on the SemanticKITTI dataset, focusing on the object classes to be identified (vehicles, pedestrians, façades, and traffic signs). Our contributions are listed below:

- Building upon the VPA-Net architecture, we extend its multimodal fusion framework beyond dense semantic segmentation by introducing a learnable global descriptor head that extracts compact scene-level embeddings directly from refined 3D features.
- We integrate a homography-based panoramic mosaicing strategy to enlarge the field of view and strengthen multimodal spatial correspondence, enabling more context-aware feature learning.
- We unify dense semantic segmentation and global place-level representation within a single end-to-end trainable framework, bridging multimodal perception and scene retrieval without modifying the core backbone structure.

The remainder of this paper is structured as follows. Section II reviews the related work. Section III presents the details of our proposed network. Section IV discusses the experimental results. Conclusions and future work are drawn in the last section.

II. Related Works

TABLE I: CLASSICAL HANDCRAFTED LOCAL AND GLOBAL FEATURE DESCRIPTORS FOR LIDAR DATA

Ref.	Type	Descriptor	Main approach	Robust to Viewpoint	Semantic	Strengths	Limitation
[23]	Local	Handcrafted	Gradient-based keypoint descriptor	keypoint-based	X	Robust to illumination changes; foundational for local matching	Originally 2D; not designed for raw 3D LiDAR; computationally expensive
[24]		Handcrafted	Gradients of LiDAR intensity for local description	local frame	X	Exploits intensity information; suitable for map-based localization	Sensitive to intensity noise; depends on sensor calibration
[25]		Handcrafted	Intensity distribution-based descriptor	Partial	X	Improves distinctiveness over geometry-only descriptors	Performance depends on intensity consistency across sessions
[26]		Handcrafted geometric	Histogram of surface normals and curvature relationships	Local frame	X	Fast; widely adopted; effective for registration	Sensitive to noise and sparse point clouds; limited discriminative power
[27]	Global	Semantic	Histogram/topological structure enriched with semantic classes	✓	✓	Robust to structural changes in urban scenes	Depends on segmentation accuracy
[28]		Topological-semantic	Graph-based topological descriptor with semantic labels	✓	✓	Robust to viewpoint variations; interpretable structure	Sensitive to semantic misclassification
[29]		Geometric	Laser-based urban structural descriptor	✓	X	Effective in structured urban environments	Limited performance in unstructured scenes
[30]		Projection-based	Polar image representation (Iris-like encoding)	✓	X	Rotation-invariant; compact representation	Sensitive to dynamic changes
[31]		Segmentation-based	Egocentric segmentation-driven structural descriptor	✓	✓ partial	Encodes structural context; improved distinctiveness	Dependent on reliable segmentation
[32]		Semantic	Extension of Scan Context with semantic labels	-	✓	High robustness to seasonal and illumination changes; efficient matching	Increased complexity compared to Scan Context
[33]		Compact geometric	Compact 3D histogram-based descriptor	-	X	Very fast; suitable for indoor mapping	Lower discriminative power in complex scenes
[34]		Statistical	Normal distribution modeling of geometric features	✓	X	Compact and noise-robust	Sensitive to major structural variations
[35]		Geometric	Structural descriptor for registration and LCD	✓	X	Robust to occlusions; useful for joint registration	Moderate computational cost
[36]		Projection-based	Double 3D-to-2D projection strategy	-	X	Higher discriminability than single projection methods	Possible loss of fine 3D information
[37]		Geometric	Enhanced Cartesian histogram representation	-	X	Lightweight; real-time capable	Sensitive to dynamic objects and structural changes

Research on 3D descriptors for place recognition and SLAM has evolved from classical handcrafted methods to learning-based and multi-modal fusion approaches. Early techniques relied on manually designed geometric features extracted from point clouds, either locally or globally, providing efficient but limited representations. More recently, deep learning methods operating solely on point clouds have demonstrated superior discriminative power by automatically learning invariant and task-optimized features. However, since LiDAR geometry alone may be insufficient in perceptually ambiguous or long-term changing environments, there is growing interest in learning descriptors from fused sensory information, particularly LiDAR–camera combinations, to exploit complementary geometric and visual cues. Motivated by this, this section reviews first the analyzes of classical geometric descriptors, then learning-based methods using only point clouds, and finally approaches based on LiDAR–camera fusion, highlighting the importance of multi-modal learned representations for robust long-term localization.

A. *LiDAR-Based Handcrafted Local and Global Descriptors*

A LiDAR-based local descriptor is a compact representation extracted from the geometric neighborhood of a 3D point in a LiDAR point cloud. It encodes local spatial structure, shape distribution, and possibly reflectance information to facilitate correspondence estimation, feature matching, or higher-level scene understanding tasks.

Table I summarizes key handcrafted local and global descriptors for LiDAR point clouds, highlighting their approaches, invariance properties, strengths, and limitations. Handcrafted methods such as Spin Images [23] and LiDAR intensity-based descriptors [24] rely on geometric and intensity gradients to capture local structure, offering robustness to illumination changes but often at high computational cost. Learning-based approaches like USIP [38] and PPF-AE [39] leverage unsupervised learning to identify stable keypoints and encode geometric relationships, providing improved repeatability and robustness but with increased complexity. Overall, while handcrafted descriptors laid the foundation for local feature extraction in 3D point clouds, learning-based methods have shown superior performance by automatically optimizing representations for specific tasks, albeit with trade-offs in computational efficiency.

While LiDAR-based global descriptor is a holistic representation computed from a full 3D LiDAR scan, encoding global geometric layout and structural patterns of the environment into a compact embedding suitable for place recognition and loop closure detection.

These descriptors range from geometric and projection-based methods to segmentation-driven approaches, each with unique advantages and challenges. For instance, semantic-enriched descriptors [27], [28] offer improved discriminability but depend heavily on accurate segmentation, while geometric descriptors [29], [35] are effective in structured environments but may struggle in unstructured scenes. Projection-based methods [30], [36] provide compact representations but can lose fine 3D details. Overall, while these handcrafted global descriptors have been foundational in LiDAR-based place recognition and localization, they often face limitations in dynamic or changing environments, motivating the shift towards learning-based and multi-modal approaches for enhanced robustness.

B. *Learning-Based LiDAR Local and Global Descriptors*

Learning-based local descriptors are feature representations automatically learned from data using neural networks, encoding the geometric and/or semantic structure of local neighborhoods within a 2D or 3D space. This strategy is divide into 5 stages, as depicted in Table II: 1) point-based methods, which directly operate on raw point clouds; 2) voxel-based methods, which discretize the point cloud into a voxel grid and apply 3D convolutions; 3) segment-based methods, which extract local segments from the point cloud and learn descriptors based on their geometric and semantic properties; 4) projection-based methods, which project the 3D point cloud onto 2D planes; and 5) graph-based methods, which represent the point cloud as a graph structure and learn descriptors based on node and edge features.

Descriptor evolution across the references reflects increasing abstraction and contextual modeling. Early point-based descriptors such as PointNetVLAD [10] and its extensions [40], [41], [42], rely on NetVLAD aggregation for global feature pooling.

Sparse convolution-based descriptors like MinkLoc3D [12] and MinkLoc3D-SI [44] introduce hierarchical 3D feature encoding. EgoNN [45] integrates attention mechanisms for enhanced global retrieval. Voxel-based descriptors such as NDT-Transformer [46] and TransLoc3D [47] employ statistical encodings and transformer architectures to capture long-range spatial dependencies. Segment-based descriptors like SSC [49], PCA-SSC [50], and PSE-Match [51] focus on geometric invariance and probabilistic similarity modeling. Song et al. [52] propose hybrid descriptors combining geometry, intensity, and topology. Projection-based descriptors including ADLNet [53], SphereVLAD [54], SphereVLAD++ [55], DeepRING [56], and BEVPlace [57] utilize CNN encoders on structured projections, incorporating multi-scale aggregation and rotation-equivariant mechanisms. Graph-based descriptors introduce a structural paradigm shift. SGPR [58] and GNN-based matching methods [59] encode semantic graphs and perform graph similarity learning. Cong et al. [60] integrate two-branch networks combining local spatial relations and graph-level topology. Box-Graph [61] and DGC Scan Context [62] further exploit structural descriptors and graph matching metrics for robust place recognition. Overall, descriptor development evolves from global feature aggregation toward relational graph modeling and topology-aware embeddings.

Feature representations vary significantly across paradigms. Point-based works such as U-Net [43] and MinkLoc3D [12] leverage sparse 3D convolutions, second-order pooling, and geometric descriptors. These methods focus on hierarchical geometric encoding. Voxel-based features in NDT-Transformer [46] rely on statistical distribution modeling (Normal Distribution Transform), while TransLoc3D [47] introduces adaptive receptive field modules and NetVLAD aggregation for enhanced contextual capture. Segment-based approaches such as SSC [49] and PCA-SSC [50] employ PCA-derived geometric features, whereas PSE-Match [51] models shapes probabilistically. Song et al. [52] integrate hybrid feature combinations including geometry and intensity. Projection-based methods like SphereVLAD [54], DeepRING [56], and BEVPlace [57] extract features via CNN backbones applied to spherical or BEV projections, often incorporating rotation invariance and multi-scale representations. Graph-based features explicitly encode nodes (e.g., semantic objects, segments, scan contexts) and edges (spatial relationships, topological adjacency). SGPR [58] builds semantic graphs, while [59] and [60] incorporate attention mechanisms and relational reasoning. Box-Graph [61] and DGC Scan Context [62] emphasize structural graph descriptors and pose-invariant relational encoding. Thus, feature design increasingly transitions from local geometric encoding toward structured relational representations.

Similarity computation strategies reflect methodological differences across paradigms. Most deep descriptor-based approaches, including [10], [12], [42], [43], [44], [47], [49], [50], [52], [53], [54], [55], [57], [62], rely on L2 or cosine similarity in embedding space for efficient large-scale retrieval. NetVLAD-based methods [10], [41], primarily use cosine similarity. SSC-based methods [49], [50], use L2 distance on PCA-derived descriptors. PSE-Match [51] employs a probabilistic similarity measure. SA-LOAM [52] is distinct in using ICP-based alignment. DeepRING [56] introduces

TABLE II: LEARNING-BASED LIDAR LOCAL AND GLOBAL DESCRIPTORS

Ref.	Class	Descriptor	Feature	Similarity
[10]	Point-based	PointNet	NetVLAD	L2 / Cosine
[40]		GNN, DG-CNN	NetVLAD	Cosine
[41]		PointNet + FlexConv	NetVLAD, SE block	Cosine
[42]		Pyramid Point Transformer	Pyramid VLAD	L2 / Cosine
[43]		U-Net	ePN, 2nd-order pooling	L2
[12]	Voxel-based	MinkLoc3D	Sparse 3D convolutions, FPN, GeM	L2
[44]		MinkLoc3D-SI	Sparse convolutions - spherical coordinates - intensity; FPN; GeM	L2
[45]		EgoNN	FPN backbone; ECA attention	Uses global descriptor retrieval
[46]		NDT-Transformer	Voxel-level statistical representation (Normal Distribution Transform); Transformer encoder;	Uses descriptor retrieval
[47]		TransLoc3D	NDT representation; Adaptive Receptive Field Module (ARFM); Transformer; NetVLAD	L2
[48]	Segment-based	SA-LOAM	Edge and planar features (LOAM style); spatial-appearance cues	ICP-based
[49]		SSC	PCA-based shape descriptors from segmented point clouds	L2 between PCA descriptors
[50]		PCA-SSC	Improved PCA descriptors with stronger geometric invariance	L2 distance
[51]		PSE-Match	Probabilistic Shape Encoding (mixture-model representation)	Probability-based similarity measure
[52]		Song et al.	Hybrid descriptor combining geometry, intensity, and local topology	L2 or cosine
[53]	Projection-based	AttDLNet	Attention-based CNN encoder on projection images; multi-scale feature extraction	L2
[54]		SphereVLAD	Spherical projection + deep CNN features + VLAD aggregation	L2 / cosine distance
[55]		SphereVLAD++	Improved SphereVLAD with multi-scale projections and enhanced CNN backbone	L2 distance
[56]		DeepRING	Circular (ring-based) LiDAR projection; CNN encoder for rotation-equivariant features	Correlation-based similarity
[57]		BEVPlace	Bird's-eye-view (BEV) projection; CNN encoder with BEV-specific geometric priors	L2 distance
[58]	Graph-based	SGPR	Builds semantic graphs from segmented point clouds	Graph matching using structural similarity metrics
[59]		GOS Match	Graph-of-semantics representation where each node = semantic instance and edges = geometric/topological relations	Graph similarity scoring with node - edge
[60]		Gong et al.	Two-tier representation: local spatial relation graph + global spatial topology; semantic and geometric cues	Coarse-to-fine graph matching;
[61]		BoxGraph	Semantic bounding box graph generated from object detectors; boxes as nodes, box relations as edges	Graph matching with similarity metrics
[62]		Deep Scan Context	Learned high-dimensional scan context descriptor; pseudo-cylindrical projection + neural embedding	L2

correlation-based similarity for rotation robustness. Voxel-based NDT-Transformer [46] emphasizes descriptor retrieval frameworks. Graph-based approaches adopt more complex structural matching strategies. SGPR [58] uses graph matching and structural similarity scoring. GNN-based matching [59] integrates graph similarity learning with node and edge-level reasoning. Cong et al. [60] apply contrastive graph matching. Box-Graph [61] and DGC Scan Context [62] employ graph similarity metrics and L2 distance over relational descriptors. Overall, similarity computation evolves from simple metric comparisons toward structured graph matching and relational reasoning, trading computational simplicity for structural robustness.

C. Camera-LiDAR fusion methods

The following analysis examines recent multimodal descriptor approaches by organizing them according to descriptor

TABLE III: CAMERA-LIDAR-FUSION DESCRIPTORS

Ref.	Descriptor Type	Primary task	Multi modal Strategy	Strengths	Limitation
[63]	Global multimodal descriptor	Place Recognition	Late fusion of LiDAR + image embeddings	Combines geometric and visual cues; strong generalization	Requires synchronized L-C; moderate computational cost
[64]			Colored structural + visual fusion	Integrates appearance and structure; robust place recognition	Slightly lower recall than top-performing multimodal methods
[65]			Cross-modal alignment - orientation voting	Orientation robustness via voting; cross-modal alignment	Sensor calibration complexity; computational overhead
[66]			Multi-scale LiDAR–camera attention fusion	Captures geometric and appearance cues; robust under viewpoint change	Accurate cross-modal calibration; attention increases computational cost
[67]			Bi-directional attentional fusion	Attentional refinement; resilient to ambiguous scenes	Higher computational cost due to attention mechanism
[68]	Semantic feature descriptor	Semantic Segmentation	Multi-modal driving fusion (RGB + LiDAR)	Accurate multimodal segmentation; integrates camera + LiDAR	Computationally heavy; dataset-specific performance
[69]			Cross-modal transformer fusion (RGB-X)	Strong cross-modal feature learning; improved semantic accuracy	Transformer complexity; high GPU memory usage
[70]			LiDAR + teacher features (cross-modal distillation)	Unsupervised knowledge distillation; improves LiDAR descriptors	Dependent on teacher model; added training complexity
[71]		Unsupervised Domain Adaptation	Multi-modal semantic adaptation	Handles domain shift; adapts features without supervision	Limited semantic precision compared to fully supervised methods
[72]			Multi-modal prior-guided cross-domain adaptation	Leverages image priors to reduce domain gap; improves target generalization	Relies on quality of pseudo-labels; sensitive to domain discrepancy
[73]		Semantic Segmentation	Cross-modal semantic attention	Focused semantic fusion; enhances point-level semantics	Requires accurate alignment between sensors
[22]			Vision assistance for LiDAR	Leverages visual context to improve LiDAR features	Needs dense camera coverage; limited generalization if camera fails
[21]			Range + image fusion	Real-time performance; robust multimodal segmentation	Requires synchronized RGB; may struggle with sparse LiDAR
[74]			Depth-aware transformer fusion	Attention-based multimodal context; strong semantic consistency	Transformer complexity; high memory usage
[75]			LiDAR + fisheye fusion	Wide field-of-view coverage; captures rich context	Needs fisheye camera; calibration-sensitive
[76]			Efficient perception fusion	Efficient fusion; perception-aware features	Focused on efficiency, may trade off accuracy in complex scenarios
[77]			Collaborative multimodal learning	Robust semantic feature learning across modalities	Higher training complexity; needs aligned multimodal data
[78]			Camera–LiDAR cross-attention with fast neighbor aggregation	Efficient local-global feature interaction; enhanced boundary precision	Attention and neighbor aggregation increase memory usage
[79]			Early–mid LiDAR–image feature fusion	Improves discriminative power of LiDAR features using visual cues	Dependent on synchronized RGB; performance drops under poor lighting
[80]			Feature-level RGB–LiDAR fusion with hierarchical encoder	Enhanced small-object segmentation; better contextual modeling	Dataset-dependent tuning; moderate computational overhead
[81]			Transformer-based multimodal fusion	Strong long-range cross-modal int.; improved semantic consistency	High GPU memory consumption; complex training
[82]			Unified multimodal encoder (optical + auxiliary modalities)	Cost-effective design; scalable multimodal integration	Designed for remote sensing; limited direct transfer to autonomous driving
[83]			Hierarchical transformer-based multimodal fusion	Multi-scale contextual modeling; strong global reasoning	Computationally intensive; large training data requirement
[84]		Semantic Scene Completion	Continuous fusion in voxelized 3D grid	Dense 3D context modeling; consistent volumetric representation	High memory usage due to voxel.; computationally heavy
[85]		3D Vehicle Detection	Semantic-guided camera–LiDAR feature augmentation	Enhances object-aware fusion; improves detection robustness	Reliable semantic supervision; added training complexity

type, primary task, fusion strategy, strengths, and limitations. By structuring the comparison column-wise, this discussion highlights the evolution from global multimodal embeddings for place recognition toward semantically rich feature representations for dense scene understanding, while also identifying common design trends, performance advantages, and practical challenges associated with multimodal perception systems. The Table III distinguishes between Global multimodal descriptors ([63]–[67]) and Semantic feature descriptors ([21], [22],[68]-[85]), reflecting two fundamentally different design objectives. Global multimodal descriptors are primarily designed for place recognition, where LiDAR and visual data are fused to generate compact embeddings suitable for large-scale retrieval. These approaches emphasize discriminative global representation, cross-modal consistency, and robustness to viewpoint or illumination changes. In contrast, semantic feature descriptors target dense scene understanding tasks such as semantic segmentation, domain adaptation, scene completion, and object detection. Rather than producing a single global vector, these methods learn structured, context-rich feature maps that enable fine-grained prediction. This categorization highlights the methodological shift from retrieval-oriented embeddings to dense multimodal perception frameworks.

The section of primary tasks is boarded from global localization to comprehensive scene interpretation. Works [63]–[67] focus on place recognition, aiming to achieve reliable loop closure detection through multimodal global embeddings. From [85] onward, the emphasis shifts toward semantic segmentation, where multimodal fusion enhances pixel- or point-level classification accuracy. Domain adaptation strategies ([72], [73]) address distribution shifts between datasets, improving generalization. More advanced tasks include semantic scene completion ([84]), which infers volumetric occupancy and semantics, and 3D vehicle detection ([85]), which strengthens object-level reasoning through feature augmentation. This progression illustrates increasing task complexity and a move from global matching toward dense and structured environment understanding. The multimodal strategies vary in fusion depth and architectural sophistication. Early approaches ([63], [79]) rely on direct LiDAR–image feature concatenation, while others introduce cross-modal alignment and orientation voting mechanisms ([65], [70], [73]) to enforce feature consistency. Attention-based fusion models ([67], [74], [78]) selectively weight complementary modality cues, improving robustness in ambiguous scenes. Transformer-based frameworks ([69], [81], [83]) capture long-range contextual dependencies across modalities, representing a significant architectural advancement. Domain adaptation techniques ([72], [73]) leverage pseudo-label guidance to reduce domain gaps. Task-specific fusion schemes, such as voxelized semantic grids ([84]) or feature augmentation for detection ([85]), tailor multimodal integration to downstream objectives. Overall, there is a clear evolution from simple feature fusion to structured cross-attention and unified encoder architectures. Across the references, multimodal approaches consistently demonstrate improved robustness and contextual understanding. Methods such as [63] and [64] effectively integrate geometry and visual structure, enhancing recognition reliability. Cross-modal alignment techniques ([65], [70]) improve orientation consistency and discriminative capacity. Attention-based and transformer-driven models ([67], [69], [81], [83]) provide stronger global reasoning and context modeling, leading to improved segmentation accuracy and semantic consistency. Domain adaptation frameworks ([72], [73]) enhance generalization to unseen environments. Scene completion ([84]) and detection frameworks ([85]) benefit from enriched object-aware fusion. In general, multimodal learning leverages complementary sensor characteristics to increase accuracy, resilience, and scene comprehension. Despite their advantages, these approaches introduce several practical constraints. Many methods ([63], [66], [76]) require accurate LiDAR–camera calibration and synchronization, which may be difficult in real-world deployments. Transformer-based and attention-heavy architectures ([69], [81], [83]) significantly increase computational cost and training complexity. Domain adaptation techniques ([72], [73]) depend on pseudo-label quality and may struggle with severe distribution shifts. Some systems require specialized hardware, such as fisheye cameras ([75]), limiting scalability. Additionally, multimodal models often demand large aligned datasets ([77], [85]) and may exhibit sensitivity to sensor degradation or adverse lighting conditions. Therefore, while multimodal fusion improves perception performance, it also raises challenges in efficiency, scalability, and deployment robustness.

In summary, prior work shows that multimodal fusion substantially improves robustness and contextual awareness in both place recognition and dense perception tasks. Nevertheless, many existing approaches either prioritize global retrieval over object-level discrimination or focus on dense prediction without explicitly learning compact and discriminative multimodal descriptors. Moreover, challenges related to efficient fusion, scalability, and generalization remain open. In this context, we propose a method centered on the generation of multimodal fused descriptors that, through learning-based strategies, enable more robust object discrimination and a deeper understanding of the surrounding environment.

III. Methodology

The methodology for generating descriptors from a learning process based on the fusion of images and point clouds is divided into several stages: The first stage corresponds to the adjustment or preprocessing of the point cloud so that it can be projected onto the panoramic image. Then, based on the model presented by [22], the point cloud and the images are fed into the network to obtain perceptual features from both modalities individually through feature fusion using attention-based models. Next, the projected point cloud is transformed into the 3D space using sparse convolutions within a refinement module to optimize the segmentation produced by both modalities. Finally, a trainable global descriptor is constructed, which transforms the multimodal 3D features into a discriminative embedding for the identification of the objects present in the scene.

A. *Panoramic Images Generation*

For comprehensive scene understanding and environmental awareness, the use of panoramic images offers significant advantages over single-camera setups, particularly in outdoor and large-scale autonomous perception scenarios. The panoramic representation of our multi-sensor system provides a 180-degree field of view, enabling the system to capture in front of the vehicle spatial context of a scene in a single frame.

Fig. 2: Panoramic Image Generation via Multi-Camera Homography Alignment and Stitching.

This holistic perception reduces blind spots and allows the model to reason about long-range spatial relationships between objects that would otherwise be distributed across multiple independent camera views. By observing the entire surrounding environment simultaneously, the system gains a more coherent representation of scene layout, object distribution, and structural continuity.

In contrast, single-camera images are inherently limited by their narrower field of view, which can restrict contextual reasoning and lead to partial observations of large structures such as buildings, intersections, or dynamic objects. This limitation may require additional multi-view fusion strategies to approximate a global understanding of the environment. Panoramic imagery, therefore, naturally supports richer contextual modeling, improved spatial completeness, and more consistent global representations of complex scenes. The process begins with feature extraction, where salient keypoints and corresponding binary descriptors are detected using the ORB [5] algorithm. ORB is selected due to its computational efficiency and robustness to rotation and illumination changes, making it suitable for real-time multi-camera applications. The input images have an initial spatial resolution of 1280 × 720 pixels, providing sufficient detail for reliable keypoint detection and matching. Following feature extraction, a keypoint matching stage is performed between overlapping image pairs. Descriptor correspondences are established using a nearest-neighbor search strategy, typically based on Hamming distance due to ORB's binary descriptors. These correspondences provide the geometric constraints necessary for estimating the transformation between views.

Next, a homography matrix is computed using the matched keypoints. To ensure robustness against incorrect correspondences (outliers), the homography is estimated using the RANSAC (Random Sample Consensus) algorithm. RANSAC iteratively selects minimal subsets of correspondences to compute candidate homographies and retains the solution that maximizes inlier consistency, thereby improving geometric accuracy. Once the homography matrix is obtained, a perspective warping step is applied. One image is projected onto the coordinate frame of the reference image using a perspective transformation, aligning the overlapping regions into a common plane. This warping compensates for viewpoint differences between cameras under the planar scene assumption. Finally, after geometric alignment and stitching, the resulting composite image undergoes cropping to remove redundant regions and boundary artifacts. The final panoramic output has a spatial resolution of 908 × 230 pixels, producing a compact, horizontally extended representation suitable for downstream perception tasks. This process is showed in Fig. 2.

B. Preprocessing of the Point Cloud

Several strategies can be employed to project 3D point clouds onto the image domain, including perspective, cylindrical [86], and spherical projections [87]. The selection of the projection model is strongly influenced by the imaging geometry, sensor configuration, and especially the field of view of the camera system. In scenarios involving panoramic images with a wide angular coverage, spherical projection offers a more consistent geometric formulation than conventional perspective projection. Unlike perspective models, which are inherently limited to narrower fields of view and may introduce severe distortions at large viewing angles, spherical parameterization maintains the angular continuity of the scene by directly mapping LiDAR points according to their azimuth and elevation angles. This property ensures a more uniform spatial correspondence between 3D points and image pixels, thereby improving the stability of cross-modal alignment and facilitating more reliable multimodal feature fusion. To perform the spherical projection as depicted in Fig. 3, we first convert the Cartesian coordinates of the LiDAR points into spherical coordinates. Each point in the point cloud is represented by its range r, azimuth angle θ, and elevation angle ϕ. These type of images are dependent on the LiDAR's design parameters. It is important to know the vertical field of view (FOV) and the number of channels of the LiDAR to determine the resolution of the projected image. The vertical FOV defines the range of elevation angles that the LiDAR can capture, while the number of channels determines how many discrete elevation angles are sampled within that FOV.

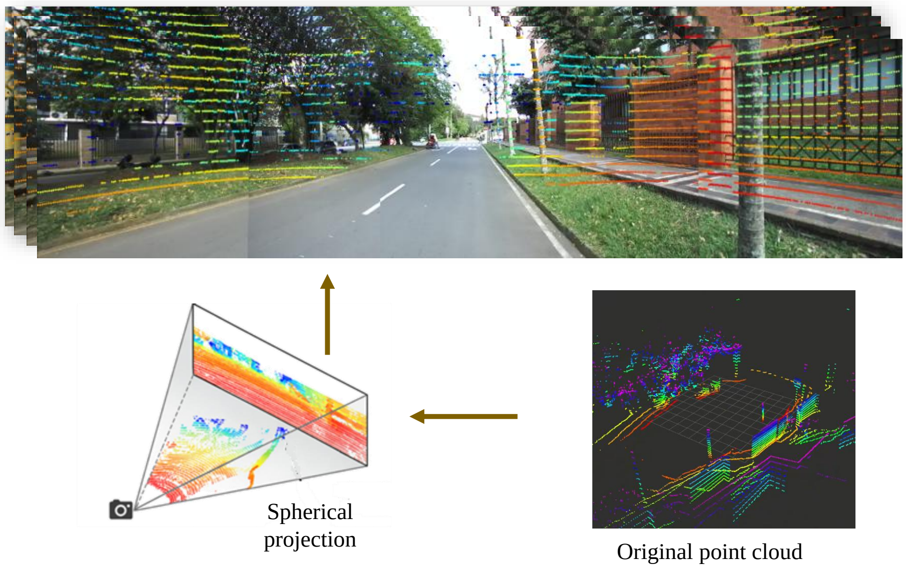

Fig. 3: Spherical Projection of LiDAR Point Cloud onto the Panoramic Image Plane.

This means that each channel corresponds to a specific elevation angle, and the point cloud can be projected onto a 2D image where each pixel corresponds to a specific azimuth and elevation angle. The horizontal resolution is determined by the azimuthal sampling, which is typically uniform across the 360-degree horizontal field of view. We comprte the depth of each point in the point cloud as the Euclidean distance from the LiDAR sensor to the point, which can be calculated using the formula:

$$r = \sqrt{x^2 + y^2 + z^2} \tag{1}$$

where x, y, and z are the Cartesian coordinates of the point. Then, we calculate the azimuth angle θ and elevation angle ϕ for each point using the following formulas:

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) \tag{2}$$

$$\theta = \sin^{-1}\left(\frac{z}{r}\right) \tag{3}$$

Afterward, each of the angles needs to be normalized to range between $[0, 1]$, and then remapped to the perspective ranges of $[0, W]$ for yaw and $[0, H]$ for pitch, where W and H are the width and height of the projected image, respectively. The remapping can be performed using the following equations:

$$u = 0.5 \cdot \frac{\phi}{\pi} \cdot W \tag{4}$$

$$v = \left(1.0 - \left(\frac{\theta + |fov_{down}|}{fov}\right)\right) \cdot H \tag{5}$$

where fov is the total vertical field of view of the LiDAR, and fov_{down} is the downward vertical field of view. The resulting coordinates (u, v) represent the pixel location in the projected image corresponding to each point in the point cloud. This process effectively transforms the 3D spatial information from the LiDAR into a 2D representation that can be easily fused with the panoramic image data for subsequent feature extraction and descriptor generation.

C. Multi-modal Semantic Segmentation

IV. Experiments and Results

V. Conclusions

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